



Hall Ticket No-

Code: S123528

Stanley College of Engineering and Technology for Women(A)**B.E.(CSE/EEE/ECE/IT/CME/AI&DS) II Semester (Backlog) Examinations May-June-2023****Mathematics-II**

Time: 3 hours

Max.Marks:60

PART-A**Note: Answer all questions (Compulsory)****5 X 2=10M**

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|---|--|-----|----|
| 1 | Write standard form of Bernoulli's differential equation of first order. | 2 M | L1 |
| 2 | Find the Particular Integral of $(D^2 + 4^2)y = e^{-4x}$ | 2 M | L1 |
| 3 | Define the Gamma function. | 2 M | L1 |
| 4 | State Cayley-Hamilton theorem. | 2 M | L1 |
| 5 | Find the Laplace Transform of t^3 | 2 M | L1 |

PART-B**Note: Answer all questions****5 X 10= 50M**

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|-----------|---|-----|----|
| 6 | a. Find the Orthogonal Trajectory of the family of curves $r^n = a^n \cos n\theta$, where θ is parameter. | 5M | L4 |
| | b. Solve $(3x^2 + 6xy^2)dx + (6x^2y + 4y^3)dy = 0$ | 5M | L3 |
| OR | | | |
| | c. Solve $(3y^2 + 4x) dx + 3xydy=0$ | 5M | L4 |
| | d. Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$ | 5M | L3 |
| 7 | a. Solve $\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 0$, given $x(0) = 0, \frac{dx}{dt}(0) = 15$ | 5M | L3 |
| | b. Solve $(D^2 - 6D + 25)y = e^{2x}$. | 5M | L3 |
| OR | | | |
| | c. Solve by the method of variation of parameters, $\frac{d^2y}{dx^2} + 4y = \tan 2x$. | 10M | L3 |
| 8 | a. Show that $\int_a^b (x-a)^m (b-x)^n dx = (b-a)^{m+n-1} \cdot \beta(m+1, n+1)$ | 5M | L3 |
| | b. Show that $\int_0^\infty \sqrt{x} e^{-x^3} dx = \frac{\sqrt{\pi}}{3}$ | 5M | L3 |
| OR | | | |
| | c. State and Prove the relation between Beta and Gamma. | 5M | L3 |
| | d. Prove that $\beta\left(m, \frac{1}{2}\right) = 2^{2m-1} \beta(m, m)$ | 5M | L3 |

- 9 a. Reduce the quadratic form into canonical form by an orthogonal transformation. Also state the nature of quadratic form

10M L2

$$8x_1^2 + 7x_2^2 + 3x_3^2 - 12x_1x_2 - x_2x_3 + 4x_3x_1$$

OR

- c. Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} .

5M L2

- d. Find the Eigen values of $\text{adj } A$ and $A^2 - 2A + I$, where $A = \begin{pmatrix} 2 & 3 & 4 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{pmatrix}$

5M L2

- 10 a. Find the Laplace transform of $t^3 e^{-3t}$.

5M L3

- b. Find the inverse Laplace transform of the $\frac{1}{s(s^2+a^2)}$ by convolution theorem.

5M L3

OR

- c. Solve, using Laplace transform method, $y'''' + 2y'' - y' - 2y = 0$ with the conditions $y(0) = y'(0) = 0$ & $y''(0) = 6$

10M L3

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Engineering Physics

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PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

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|---|---------|
| 1 Classify Fraunhofer and Fresnel diffraction patterns. | 2 M BL2 |
| 2 List any four applications of Lasers. | 2 M BL1 |
| 3 Define Unit cell and Bravais lattice. | 2 M BL1 |
| 4 What is de-broglie's concept of matter waves? | 2 M BL1 |
| 5 What is Biot-savart's law? | 2 M BL1 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | |
|--|--------|
| 6 a. Explain the formation of Newton's rings. | 4M BL4 |
| b. Discuss the conditions for maxima and minima for diffraction at single slit. | 6M BL6 |
| OR | |
| c. Classify the interference of light by division of amplitude and by division of wavefront. | 4M BL2 |
| d. Explain the distribution of intensity due to N slit grating. | 6M BL2 |
| 7 a. Explain the construction and working of He-Ne laser. | 6M BL2 |
| b. Explain the concept of numerical aperture in fiber optics. | 4M BL2 |
| OR | |
| c. Explain the characteristics of lasers. | 4M BL2 |
| d. Explain Double Crucible method to produce optical fiber. | 6M BL2 |
| 8 a. Explain Bravais lattices in crystal systems. | 6M BL2 |
| b. Explain line defects and plane defects that occur in crystals. | 4M BL2 |
| OR | |
| c. Develop an expression for the concentration of Schottky defects in ionic crystals. | 6M BL3 |
| d. How Miller indices are used to identify the orientation of planes in crystals? | 4M BL1 |
| 9 a. Develop Schrodinger time independent wave equation. | 6M BL3 |
| b. Explain electric polarization in dielectrics. | 4M BL1 |
| OR | |
| c. Develop an expression for wavefunction associated with a particle in 1-D box. | 6M BL3 |
| d. Explain Gauss law of electricity. | 4M BL2 |
| 10 a. Build an equation for plane electromagnetic wave in free space. | 6M BL3 |
| b. Explain magnetic flux density and give Maxwell's equation | 4M BL2 |
| OR | |
| c. Develop Maxwell's equation in integral and differential form for Ampere's law. | 6M BL3 |
| d. Explain Faraday's law of electromagnetic induction. | 4M BL2 |
